Chapter 1

1. Define software? What are different natures of the software that are different than hardware?

2. Explain failure curve of hardware (bathtub curve) and software

2. Define software myths. Write down different software myths of customer, manager and developers with reality.

3. What is software crisis? Write down the causes and remedies of software crisis.

4. What is project? Explain about Four Ps concept used in software engineering.

5. Define measure, metric and indicator. Calculate LOC and FP numerical from copy.

6. Write down the demerits of LOC. How do function point overcome limitations of LOC explain?

7. Define empirical cost estimation technique. Explain about COCO model.

8. Define risk. Explain different types of software risk.

9. Write down different risk management steps. Explain each steps in detail.

10. What is RMMM plan. Develop RMMM plan for staff turnover in any company.

12. What is risk table? How does risk table help to minimize the risks. Explain.

13. Define software ethics. Why ethics is essential for software engineers.

Chapter 2

1. Define software process. Explain different phases of software development life cycle (SDLC).

2. Why should we follow software process while developing software? Write down the pros and cons of Waterfall model.

3. If the customer requirement is vague which model do you prefer and why?

4. For every model (Waterfall, incremental, RAD, prototype, spiral model) write down its merits demerits and when to use this model.

5. What are the pros and cons of agile method over traditional method? Write down.

6. Define agility. Explain SCRUM process in detail with necessary diagrams.

7. Write down four values of extreme programming (XP). Explain XP process in details with necessary diagrams.

8. Define terminology aspects, join points, point cuts, cross cutting concerns. What are the advantages of aspect oriented software engineering.

Chapter 3

1. Define functional and non-functional requirements. List out functional and non-functional requirements for library management system.

2. Why do we need analysis principles? Explain some requirement analysis principles.

3. What is context model? Create context model for library system.

4. Explain how data flow diagram DFD helps to model functionality of a system. Explain with necessary diagrams.

5. What is data modeling? How can we model data with the help of Entity relationship diagram? Explain with suitable example.

6. What is behavior modeling? Explain how state transition diagram, sequence diagram and activity diagrams can used to depict behavior of a system. Explain (to solve this question student need to draw all the diagrams separately taking an example of each and explain).

7. Define requirement engineering. Explain different stages of the requirement engineering process.

8. What is requirement elicitation? Explain different requirement elicitation process (interview, ethnography, user stories and scenarios).

9. Why do we need requirement validation? Explain different types of checks during requirement validation

10. Define object oriented analysis. Explain how use cases are used for analysis purpose. (Draw use case for any system)

Chapter 4

1. Differentiate between analysis and design. Explain 6 design principles.

2. Good design leads to good code. Justify. Explain design process.

3. What is architectural design? Explain in details about Layered, Repository, Client Server, Pipe and Filter with necessary diagram.

4. What is design pattern. Write down the importance of design pattern. Explain observer pattern and represent it in UML.

5. Define security by design. How can we implement security in each phases of SDLC explain.

6. Define embedded system. Explain how we can design embedded software system.

7. Draw class diagram for Library management system showing different kind of relationship and modality.

8. Draw architecture diagram of Library management system.

Chapter 5

1. Define testing. Explain importance of software testing.

2. What is verification and validation? Why verification and validation is important explain with suitable example.

3. What is development testing? Explain different stages of development testing.

4. What is release testing? Explain different stages of release testing.

5. What is user testing? Explain different stages of unit testing.

6. Define test case. Explain why test cases are important.

7. Define cyclomatic complexity V(G). Write down steps for basis path testing.

8. Calculation of cyclomatic complexity of given source code. Refer class copy.

9. What is equivalent partitioning? What is different consideration during creating equivalent classes?

10. Explain about boundary value analysis testing. For numerical consider class copy.

11. Explain software evolution process with necessary diagram.

12. What are fundamental sources of software change? Explain relation between software change and evolution.

13 Define software maitenance.Why do we need software maintenance? Describe different types of software maintenance.

14. Define reengineering. Why reengineering is important. Explain different activities involved in software reengineering process.

Chapter 6

1. Define Quality. Write down different quality attributes of software.

2. Define software reliability and availability. Write down the relationship between reliability and availability.

3. Why do we need software standards? Difference between ISO and CMMI

4. Explain different levels of CMMI and also mention its Key process area (KPA) of CMMI

5. What review and inspection. Explain review process in details.

6. Explain how review and inspection helps to ensure software quality.

7. Explain the key aspect of process and product standard.

Chapter 7

1. Define configuration management. What are different activities during configuration management?

2. Why change is required in software. Explain change control/management with necessary diagrams and steps.

3. Define code line and base line. Explain version management process with necessary diagram.

4. What is system building? Explain system building process in detail.

6. What are the problems of system building of the large software project?

7. Define system release. Explain different steps involved in system release.

Chapter 8

1. What is the importance of software reuse .Explain?

2. Why do we need application framework? Explain about MVC framework.

3. What are the uses of AI software engineering? Write the merits and demerits of using AI in software development.

4. Explain the uses of cloud platform during software development.

5. What are the merits and demerits of using platform for software development? Explain

Note:

This sample question for only assessment.